Unit 30: Inverse Functions and Their Derivatives — Full Solutions

Warm-up

1. $f(x) = 2x + 3$

$y = 2x + 3$. Swap: $x = 2y + 3$. Solve: $2y = x - 3 \Rightarrow y = \dfrac{x-3}{2}$.

$f^{-1}(x) = \dfrac{x-3}{2}$.

2. $f(x) = 5x - 2$

$y = 5x - 2$. Swap: $x = 5y - 2$. Solve: $5y = x + 2 \Rightarrow y = \dfrac{x+2}{5}$.

$f^{-1}(x) = \dfrac{x+2}{5}$.

3. $f(x) = 4x$

$y = 4x$. Swap: $x = 4y$. Solve: $y = \dfrac{x}{4}$.

$f^{-1}(x) = \dfrac{x}{4}$.

4. $f(x) = \dfrac{x}{2} + 1$

$y = \dfrac{x}{2} + 1$. Swap: $x = \dfrac{y}{2} + 1$. Solve: $x - 1 = \dfrac{y}{2} \Rightarrow y = 2(x-1) = 2x - 2$.

$f^{-1}(x) = 2x - 2$.

5. Verify $f(x) = 3x - 6$ and $g(x) = \dfrac{x+6}{3}$ are inverses.

$$f(g(x)) = 3\left(\frac{x+6}{3}\right) - 6 = (x+6) - 6 = x$$

Confirmed: $g(x) = f^{-1}(x)$.

6. $f(x) = -2x + 7$

$y = -2x + 7$. Swap: $x = -2y + 7$. Solve: $2y = 7 - x \Rightarrow y = \dfrac{7-x}{2}$.

$f^{-1}(x) = \dfrac{7-x}{2}$.

Standard

7. $f(x) = x^3$

$y = x^3$. Swap: $x = y^3$. Solve: $y = \sqrt[3]{x}$.

$f^{-1}(x) = \sqrt[3]{x}$.

8. $f(x) = \sqrt{x-2}$, $x \geq 2$

$y = \sqrt{x-2}$. Swap: $x = \sqrt{y-2}$. Solve: $x^2 = y-2 \Rightarrow y = x^2+2$.

$f^{-1}(x) = x^2+2$, with domain $x \geq 0$ (matching the range of the original function, since $\sqrt{x-2} \geq 0$ always).

9. $f(x) = x^2+3,\ x \geq 0$

$y = x^2+3$. Swap: $x = y^2+3$. Solve: $y^2 = x-3 \Rightarrow y = \sqrt{x-3}$ (taking the positive root, since the original domain restricts to $x \geq 0$).

$f^{-1}(x) = \sqrt{x-3}$, with domain $x \geq 3$.

10. $f(x) = 2x+1$

(a) $y = 2x+1$. Swap: $x = 2y+1$. Solve: $y = \dfrac{x-1}{2}$. So $f^{-1}(x) = \dfrac{x-1}{2}$, and

$$(f^{-1})'(x) = \frac{1}{2}$$

(b) $f'(x) = 2$. Using the formula:

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))} = \frac{1}{2}$$

(since $f'(x) = 2$ regardless of the input). Both methods agree: $(f^{-1})'(x) = \dfrac{1}{2}$.

11. $f(x) = x^3$, find $(f^{-1})'(8)$.

First find $f^{-1}(8)$: we need x such that $x^3 = 8$, so $x = 2$.

$f'(x) = 3x^2$, so $f'(2) = 3(4) = 12$.

$$(f^{-1})'(8) = \frac{1}{f'(2)} = \frac{1}{12}$$

Answer: $\dfrac{1}{12}$.

12. $f(x) = x^2+1,\ x \geq 0,\ f(2) = 5$.

Since $f(2) = 5$, we know $f^{-1}(5) = 2$.

$f'(x) = 2x$, so $f'(2) = 4$.

$$(f^{-1})'(5) = \frac{1}{f'(2)} = \frac{1}{4}$$

Answer: $\dfrac{1}{4}$.

Challenge

13. $f(x) = 6x - 1$

$y = 6x - 1$. Swap: $x = 6y - 1$. Solve: $6y = x + 1 \Rightarrow y = \dfrac{x+1}{6}$.

$f^{-1}(x) = \dfrac{x+1}{6}$.

14. $f(x) = x^3 + x^2 + 6x + 1$, $x \geq 0$, find $(f^{-1})'(1)$ where $a = 1 = f(0)$.

Since $f(0) = 0 + 0 + 0 + 1 = 1$, we know $f^{-1}(1) = 0$.

$f'(x) = 3x^2 + 2x + 6$, so $f'(0) = 0 + 0 + 6 = 6$.

$$(f^{-1})'(1) = \frac{1}{f'(0)} = \frac{1}{6}$$

Answer: $\dfrac{1}{6}$.

15. $f(x) = x^5 + 2x + 1$

Why explicit inversion is hard: this is a degree-5 (quintic) polynomial. Unlike quadratics, cubics, or quartics, there is no general algebraic formula (using only basic operations and roots) for solving a general fifth-degree equation for x in terms of y — this is a famous result in algebra. So writing down $f^{-1}(x)$ explicitly isn't practical here, which is exactly why the derivative formula (which sidesteps needing the explicit inverse) is so valuable.

Since $f(1) = 1 + 2 + 1 = 4$, we know $f^{-1}(4) = 1$.

$f'(x) = 5x^4 + 2$, so $f'(1) = 5 + 2 = 7$.

$$(f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{7}$$

Answer: $\dfrac{1}{7}$.

16. $f(x) = x^3 + x$

Checking one-to-one: $f'(x) = 3x^2 + 1$. Since $3x^2 \geq 0$ always, $f'(x) \geq 1 > 0$ for every x — meaning f is strictly increasing everywhere, so it's one-to-one and has a valid inverse.

Since $f(1) = 1 + 1 = 2$, we know $f^{-1}(2) = 1$.

$f'(1) = 3(1) + 1 = 4$.

$$(f^{-1})'(2) = \frac{1}{f'(1)} = \frac{1}{4}$$

Answer: $\dfrac{1}{4}$.

17. $f(x) = 2x^5 + x^3 + 1$, $f(1) = 4$.

Since $f(1) = 2+1+1 = 4$, we know $f^{-1}(4) = 1$.

$f'(x) = 10x^4 + 3x^2$, so $f'(1) = 10+3 = 13$.

$$(f^{-1})'(4) = \frac{1}{f'(1)} = \frac{1}{13}$$

Answer: $\frac{1}{13}$.

Applied

18. $f(C) = 1.8C + 32$

$F = 1.8C + 32$. Solve for C: $F - 32 = 1.8C \Rightarrow C = \frac{F-32}{1.8}$.

$f^{-1}(F) = \frac{F-32}{1.8}$.

Convert $98.6°F$:

$$C = \frac{98.6-32}{1.8} = \frac{66.6}{1.8} = 37°C$$

Answer: $98.6°F = 37°C$.

19. $C(x) = 50x + 200$

Solve for x: $\text{cost} - 200 = 50x \Rightarrow x = \frac{\text{cost}-200}{50}$.

Inverse function: $x = \frac{\text{cost}-200}{50}$.

For a budget of \$1200:

$$x = \frac{1200-200}{50} = \frac{1000}{50} = 20$$

Answer: 20 units can be produced with a \$1200 budget.

20. $p(q) = 100 - 2q$

Solve for q: $p = 100 - 2q \Rightarrow 2q = 100 - p \Rightarrow q = \frac{100-p}{2} = 50 - \frac{p}{2}$.

$q(p) = 50 - \frac{p}{2}$. This inverse function tells you the quantity consumers will demand at a given price p — it's the same relationship as the original demand function, just viewed with price as the input and quantity as the output instead of the other way around.

21. $h(x) = x^3 + 2x$, $x \geq 0$, $h(2) = 12$.

Since $h(2) = 8+4 = 12$, we know $h^{-1}(12) = 2$.

$h'(x) = 3x^2 + 2$, so $h'(2) = 12+2 = 14$.

$$(h^{-1})'(12) = \frac{1}{h'(2)} = \frac{1}{14}$$

Answer: $(h^{-1})'(12) = \dfrac{1}{14}$. This means that at the point where the rocket has burned 2 tons of fuel and reached 12 meters, roughly $\dfrac{1}{14}$ of a ton of additional fuel is needed to climb one more meter at that instant — the local "fuel cost per meter of altitude" at that specific point in the flight.